Activity

Using the given Table I, find the pattern, answer the following questions and colour the values in the given Table II. One is done for you.

Table I

Term(n)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	...
F(n)	1	1	2	3	5	8	13	21	34	55	89	144	233	377	610	

1. Where are the even Fibonacci Numbers? Colour both the term n and where **F(n) is even** in **yellow.** Do you find any pattern? Every **Third** Fibonacci number is **a multiple of 2(even).**
2. Where there are Fibonacci numbers which are **multiple of 3 ?** Colour both the term **n and** where **F(n) is multiple of 3** in **red.**

   Write down the pattern you find

   Every ……Fibonacci number is …….
3. What about the **multiple of 5**? Colour both the term **n** and where **F(n) is multiple of 5** in **blue.** Write down the pattern you find.

   Every ……i.e. ……
4. What about the **multiple of 8?** Colour both the term **n** where **F(n) is multiple of 8** in **green.** Write down the pattern you find.

   Every ……i.e. ……

Table II

Term(n)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	...
F(n)	1	1	2	3	5	8	13	21	34	55	89	144	233	377	610	

From the above activity, we conclude that
Every Fibonacci number is a factor of (a term number of) Fibonacci numbers in miltiples.

	n	3	4	5	6	7	8	9	10	11	12	13	14	15	...
	F(n)	2	3	5	8	13	21	34	55	89	144	233	377	610	...
Factors	2=F(3)	✓	×	×	✓	×	×	✓	×	×	✓	×	×	✓	Every 3rd Fib number
	3=F(4)	×	✓	×	×	×	✓	×	×	×	✓	×	×	×	Every 4th Fib number
	5=F(5)	×	×	✓	×	×	×	×	✓	×	×	×	×	✓	Every 5th Fib number
	8=F(6)	×	×	×	✓	×	×	×	×	×	✓	×	×	×	Every 6th Fib number
	F(k)														Every Kth Fib number

From the above table, we get a general rule as **Every k^{th} Fibonacci number is a multiple of F(k).**

8th_Maths_English Medium_Chapter 7.indd 250 03-12-2022 15:22:05